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PIZZA SALES DATA ANALYSIS

ABOUT DATA

"This project analyzes pizza sales data using PostgreSQL. It involves designing a relational database with 4 tables — pizza_types, pizzas, orders, and order_details — importing real CSV data, and writing SQL queries to extract key business insights such as total orders, peak hours, revenue by pizza type, and category-wise distribution."

About us

I am Chandra Prakash, a Computer Science student with a passion for data analysis and database management. This project reflects my ability to design relational databases, handle real-world datasets, and write SQL queries to extract meaningful business insights using PostgreSQL."

Calculate the total revenue generated from pizza sales.

```sql
SELECT
     sum(order_details.quantity*pizzas.price) AS Total_sal
FROM order_details join pizzas
    ON pizzas.pizza_id = order_details.pizza_id ;
```

	total_sales numeric
1	817860.05

Identify the most common pizza size ordered.

```
SELECT pizzas.size ,COUNT(order_details.order_details_id) AS Order_count
FROM pizzas JOIN order_details
ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size ORDER BY Order_count DESC;
```

	size character varying (5)	order_count bigint
1	L	18526
2	M	15385
3	S	14137
4	XL	544
5	XXL	28

List the top 5 most ordered pizza types along with their quantities.

```
SELECT pizza_types.name,
sum(order_details.quantity)AS Quantity
from pizza_types join pizzas
on pizza_types.pizza_type_id =pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.name order by  Quantity desc limit 5;
```

	name character varying (100)	quantity bigint
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Piz...	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371

Join the necessary tables to find the total quantity of each pizza category ordered

```
select pizza_types.category,
sum(order_details.quantity)as quantity
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category order by quantity desc;
```

	category character varying (50)	quantity bigint
1	Classic	14888
2	Supreme	11987
3	Veggie	11649
4	Chicken	11050

Join relevant tables to find the category-wise distribution of pizzas.

```
select category, count(name) from pizza_types
group by category;
```

	category character varying (50)	count bigint
1	Supreme	9
2	Chicken	6
3	Classic	8
4	Veggie	9

Determine the top 3 most ordered pizza types based on revenue.

```
select pizza_types.name,
sum(order_details.quantity * pizzas.price)as revenue
from pizza_types join pizzas
on pizzas.pizza_type_id = pizza_types.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.name order by revenue desc limit 3;
```

	name character varying (100)	revenue numeric
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Piz...	42768.00
3	The California Chicken Piz...	41409.50

Calculate the percentage contribution of each pizza type to total revenue.

```
select pizza_types.category,
round (sum(order_details.quantity*pizzas.price) / (SELECT
ROUND(SUM(order_details. quantity *pizzas.price),2) AS total_sales
FROM order_details
JOIN pizzas ON pizzas.pizza_id = order_details.pizza_id) *100,2) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types. category order by revenue desc;
```

	category character varying (50)	revenue numeric
1	Classic	26.91
2	Supreme	25.46
3	Chicken	23.96
4	Veggie	23.68

Analyze the cumulative revenue generated over time.

```
select date,
sum(revenue) over(order by date)as cum_revenue
from
(select orders.date,
sum(order_details.quantity * pizzas.price)as revenue
from order_details join pizzas
on order_details.pizza_id = pizzas.pizza_id
join orders
on orders.order_id = order_details.order_id
group by orders.date) as sales;
```

	date date	cum_revenue numeric
1	2015-01-...	2713.85
2	2015-01-...	5445.75
3	2015-01-...	8108.15
4	2015-01-...	9863.60
5	2015-01-...	11929.55
6	2015-01-...	14358.50
7	2015-01-...	16560.70
8	2015-01-...	19399.05
9	2015-01-...	21526.40
10	2015-01-...	23990.35
11	2015-01-...	25862.65
12	2015-01-...	27781.70
13	2015-01-...	29831.30
14	2015-01-...	32358.70
15	2015-01-...	34343.50
16	2015-01-...	36937.65
17	2015-01-...	39001.75
18	2015-01-...	40978.60

Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
select name ,revenue from
(select category,name , revenue,
rank() over(partition by category order by revenue desc) as rn
from (select pizza_types.category, pizza_types.name,
sum (order_details.quantity * pizzas.price) as revenue
from pizza_types join pizzas
on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details
on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as a) as b
where rn <=3;
```

	name character varying (100)	revenue numeric
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Piz...	42768.00
3	The California Chicken Piz...	41409.50
4	The Classic Deluxe Pizza	38180.50
5	The Hawaiian Pizza	32273.25
6	The Pepperoni Pizza	30161.75
7	The Spicy Italian Pizza	34831.25
8	The Italian Supreme Pizza	33476.75
9	The Sicilian Pizza	30940.50
10	The Four Cheese Pizza	32265.70
11	The Mexicana Pizza	26780.75
12	The Five Cheese Pizza	26066.50

Group the orders by date and calculate the average number of pizzas ordered per day.

```
select round( avg (quantity),0) as average_pizza_order_per_day
from
(select orders.date, sum(order_details.quantity) as quantity
from orders join order_details
on orders.order_id = order_details.order_id
group by orders.date) as order_quantity;
```

	average_pizza_order_per_day numeric
1	138

Determine the distribution of orders by hour of the day.

```
SELECT EXTRACT(HOUR FROM time) AS hour_of_day,
       COUNT(order_id) AS total_orders
FROM orders
GROUP BY hour_of_day
ORDER BY hour_of_day ASC;
```

	hour_of_day numeric	total_orders bigint
1	9	1
2	10	8
3	11	1231
4	12	2520
5	13	2455
6	14	1472
7	15	1468
8	16	1920
9	17	2336
10	18	2399
11	19	2009
12	20	1642
13	21	1198
14	22	663
15	23	28

Join relevant tables to find the category-wise distribution of pizzas..

```
select category, count(name) from pizza_types
group by category;
```

	category character varying (50)	count bigint
1	Supreme	9
2	Chicken	6
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4	Veggie	9

THANK YOU